One-loop shift charge of Boussinesq solitons: centroid identity, mean-profile response, and the unresolved two-body coefficient

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4 November 2026

Abstract

(c) The exhibited freedom $c(P_1M_2 - P_2M_1)$ has one coefficient and admits a rank-zero defect consistency test, but the cited package does not classify the entire continuum ambiguity. The centroid identity follows from the canonical continuity equation, subject to its domain and boundary limits. It does not justify holding the kink height fixed while moving along the quantum one-soliton dispersion branch.

(c) A fresh continuum-mode calculation gives a finite conditional mean-profile shift charge, including the finite quadratic counterterm. Its direct momentum insertion agrees with the canonical derivative of the audited one-body trace. At fixed momentum the corresponding kink-height correction is nonzero; its contribution cancels the apparent leading rational term from changing the shift charge alone. The separate, already-known Darboux recoil reproduces Toda's $2(r - r^{-1})/\pi$, but is not a constant times $P \wedge M$. **(d) Conclusion (C):** neither c_{bad} nor $c_{\mathrm{good}}(\lambda)$ is determined. The proposed identification fails as a complete mechanism; the boundary-compatible projected expectation, two-body determinant and common quantum frame remain missing.

1 Scope, evidence and the freedom actually established

Labels mean literature proof (a), literature heuristic (b), project derivation or computation (c), and an unestablished physical identification (d). No external full text or abstract was newly read and no external reference identifier is asserted. Internal sources are listed at the end. Every new displayed (c) formula is covered by the named scripts, located in `phase2/scripts/`; their accompanying JSON files retain the numerical results and explicit nulls. Conditional mathematical results are not promoted to constructed quantum observables.

(c) The fixed good convention is retained, with the bad equation used only for formal continuation and Toda validation. README Status, the master report including its ranked programme, the named packages, their relevant audits and the errata were consulted. The frame and determinant qualifications override package interpretations. Neither `audit_defect_bootstrap.tex` nor `audit_good_renorm.tex` existed when checked during this work; a reference to the latter in another audit is not evidence of its contents. No old script was executed or imported; no git operation or existing-file edit was made. No renormalisation condition is selected for the PI.

Independent structure check (c) (`shiftcharge_algebra.py`). Write $Q_{12} = P_1M_2 - P_2M_1$. In the displayed two-function ansatz,

$$F_{12} = AP_1M_2 + BM_1P_2, \qquad F_{12} + F_{21} = (A + B)(P_1M_2 + P_2M_1). \tag{1}$$

Thus antisymmetry leaves one coefficient, $A = -B = c$. All mixed 3×3 minors vanish, while mixed 2×2 minors need not vanish. This is a rank-two example, not an exhaustive classification. In particular, `defect_bootstrap.tex` explicitly retains additional unknown U_{12}. The stronger premise "the full phase is known up to one number" is not established by that package or the audits.

(c) For a good KP chord with endpoints a, b,

$$k = a - b, \quad v = \sqrt{3}(a+b), \quad a^2 + ab + b^2 = \tfrac{1}{4}, \qquad M = m(a) - m(b), \quad P = p(a) - p(b), \tag{2}$$

$$m(z) = 2z, \qquad p(z) = 2\sqrt{3}z^4 - z^2/\sqrt{3} - \sqrt{3}/36. \tag{3}$$

Polynomial reduction verifies the charge primitives independently. They telescope on a triangle. For a spectator t, the lift $W_z(t) = c[P_t m(z) - M_t p(z)]$ satisfies $W_b - W_a = -F(t, a \to b)$, with the orientation reversed if the chord is reversed. This supplies a rank-zero witness for the homogeneous cycle constraint; scalar transmission commutativity adds no rank. It does not construct a quantum absorption vertex. The corrected Toda characteristic source and its axis data, discussed in `twobody_toda.tex` and its audit, are additional dynamical inputs, not continuum bootstrap consequences.

2 What changes at fixed physical momentum

Conventions and inversion (c) (`shiftcharge_algebra.py`). For the good branch put

$$k = \sin\alpha, \quad v = \epsilon\sqrt{1-k^2}, \quad M_0 = 2k, \quad \Pi_0 = -2kv, \quad P_0 = \tfrac{2}{3}vk^3, \tag{4}$$

$$E_0 = \tfrac{2}{15}k^3(5 - 4k^2), \qquad \mathrm{d}E_0 = v\,\mathrm{d}P_0. \tag{5}$$

Take a label a away from $P_0'(a) = 0$ and write $P = P_0 + \hbar p$, $\Pi = \Pi_0 + \hbar q$, $M = M_0 + \hbar m$. At fixed physical P, the argument changes by $-\hbar p/P_0'$; hence

$$\bar{q} = q - \frac{\Pi_0'}{P_0'}p, \qquad \bar{m} = m - \frac{M_0'}{P_0'}p. \tag{6}$$

The bars are essential: fixed classical label, fixed velocity, fixed kink height and fixed physical momentum are different operations.

Topological term and frame subtraction (c) (`shiftcharge_algebra.py`). In the specified chamber-reversing convention the tree phase contains

$$T_0 = -\tfrac{2}{3}(\Pi_{0,1}M_{0,2} - \Pi_{0,2}M_{0,1}), \qquad S_{\mathrm{cb}} = \tfrac{1}{2}(\Pi_1 M_2 + \Pi_2 M_1), \quad \delta^{\mathrm{raw}} = \delta + S_{\mathrm{cb}}. \tag{7}$$

Replacing its charge arguments gives the following *part* of the order-one exponent:

$$T_1|_P = -\tfrac{2}{3}[\bar{q}_1 M_{0,2} + \Pi_{0,1}\bar{m}_2 - \bar{q}_2 M_{0,1} - \Pi_{0,2}\bar{m}_1], \tag{8}$$

$$(S_{\mathrm{cb}})_1|_P = \tfrac{1}{2}[\bar{q}_1 M_{0,2} + \Pi_{0,1}\bar{m}_2 + \bar{q}_2 M_{0,1} + \Pi_{0,2}\bar{m}_1]. \tag{9}$$

The latter must be subtracted from a raw fixed-chamber answer; it is symmetric and must not be counted again as an antisymmetric phase. **(d)** Substitution into T_0 alone is not a derivation of the full quantum phase: the dynamical term, quantum delays and state map can also change.

(c) If $\bar{m} = 0$, a sufficient condition for $T_1 = cQ_{12}$ is

$$\bar{q}(P) = \kappa P + aM_0(P), \qquad c = -\tfrac{2}{3}\kappa, \tag{10}$$

with a common constant a on the domain. It is also necessary if equality holds for every pair with nonzero M_0: divide the wedge equality by $M_{0,1}M_{0,2}$ and fix one argument. Thus proportionality to P is sufficient but not necessary; a multiple of M is invisible. A generic $\bar{q}$ instead gives a different rank-two wedge. With nonzero $\bar{m}$, the full condition is the wedge equation (8), not (10). The coefficient $-2/3$ cannot be replaced by $+1$ in this convention.

Darboux frame (c) (`shiftcharge_algebra.py`, `shiftcharge_calibration.py`). With $\gamma_0 = \mathrm{d}\Pi_0/\mathrm{d}P_0$, the corrected coefficient expressed in physical momentum is $\gamma = \gamma_0 + \hbar\partial_P\bar{q}$. If d_j denotes the unknown correction to the physical spatial delay, the two Darboux shifts change by

$$\delta\Delta X_1' = d_1 + (\partial_{P_1}\bar{q}_1)M_{0,2} + \gamma_{0,1}\bar{m}_2, \qquad \delta\Delta X_2' = d_2 - (\partial_{P_2}\bar{q}_2)M_{0,1} - \gamma_{0,2}\bar{m}_1. \tag{11}$$

Keeping $d_j = \bar{m}_j = 0$ arbitrarily is generally not integrable: the curl of the prospective phase derivatives is $-\bar{q}_1'M_2' - \bar{q}_2'M_1'$, which need not vanish. Thus a charge correction does not by itself supply a phase generator. Separately, conversion of an existing label-frame phase gives

$$R[p] = -\sum_j p_j\partial_{P_j}\delta_0 = \sum_j p_j\Delta X_j' = H[p] + D[p], \qquad D[p] = p_1\gamma_{0,1}M_{0,2} - p_2\gamma_{0,2}M_{0,1}. \tag{12}$$

This is not an extra term to add after already including the same argument shift in (6). The derivative of the full classical phase, rather than of its topological part alone, is involved.

3 Centroid identity and the kink-height sector

Local identity and boundary limit (c) (`shiftcharge_algebra.py`). The canonical Hamiltonian, including $\hbar\lambda\int u^2$, has $\dot{\phi} = \pi$ and $u_t = \pi_x$. For a smooth compact cutoff f_R define

$$B_R = \int x f_R(x)u(x)\mathrm{d}x, \qquad \dot{B}_R = -\int [f_R(x) + xf_R'(x)]\pi(x)\mathrm{d}x. \tag{13}$$

Equivalently on a finite interval, $\partial_t\int_{-R}^{R} xu\mathrm{d}x = [x\pi]_{-R}^{R} - \int_{-R}^{R}\pi\mathrm{d}x$. If the smeared limits exist and the last shell/boundary term vanishes, $\dot{B} = -\Pi$ and $\dot{M} = [\pi]_{-\infty}^{\infty} = 0$. In a nonzero fixed-height sector, M is a c-number and

$$X_{\mathrm{cm}} = B/M, \qquad \Pi = -M\dot{X}_{\mathrm{cm}}. \tag{14}$$

There is no ordering correction to this linear continuity identity. More generally the equal-time field algebra makes M commute with B; an inverse of M still needs a nonzero-mass domain. Momentum eigenstates on the line are delocalized: a localized state or a regulated matrix-element limit is needed to define its centroid. Classical kink tails do not spoil (13): π is localized, although ϕ is not. **(d)** The full quantum soliton Hilbert space and these limiting domains are not constructed by the repository's determinant packages.

Why dispersion is not sufficient (c) /(d) (`shiftcharge_algebra.py`). **(c)** If a soliton wavepacket has centroid velocity equal to its group velocity, and if its branch energy is $E(P) = E_0(P) + \hbar F(P)$, then its first-order charge relation is

$$\bar{q} = -M_0\partial_P F - v_0\bar{m}. \tag{15}$$

The group-velocity/state identification is **(d)** until supplied with the appropriate quantum sector and wavepacket. In particular, a derivative in a fixed-M sector is not automatically the derivative of a family whose $M(P)$ changes. Even granting the branch velocity identification, omitting $\bar{m}$ from (15) is an additional assumption.

(c) With fixed potential endpoints, $M = \phi(+\infty) - \phi(-\infty)$ has no perturbative correction within that sector: decaying fluctuations integrate to zero. This protects the endpoint definition, not the classical relation $M_0(P_0) = 2k(P_0)$. The quantum family can have a shifted relation between height and momentum. Already the exactly solvable elastic deformation at fixed P gives

$$(\bar{m})_{\mathrm{ct}} = -\frac{2\lambda k}{3-4k^2}, \qquad (\bar{q})_{\mathrm{ct}} = -\frac{4\lambda kv}{3-4k^2}. \tag{16}$$

These follow by solving $P = 2k^3\sqrt{1-k^2+2\hbar\lambda}/3$ to first order, and independently from the velocity-labelled mean response below. They demonstrate why exact topological conservation is not an argument for $\bar{m} = 0$ along the fixed-P comparison.

4 Independent continuum-mode insertion of the shift charge

Precisely what is computed (c) /(d) (`shiftcharge_modes.py`). **(c)** This section performs a new calculation of the continuum tadpole and the linear charge insertion on a stationary, light, good soliton, with the same free-vacuum normal ordering as the audited trace. It does not differentiate the dispersion to obtain the answer. The final momentum result is then compared with that derivative. **(d)** Completeness of the nonzero-frequency basis in the constrained quantum domain, its time-dependent projection and the finite collective measure are not established; the result is conditional on the scattering-continuum prescription. No claim to the fully projected physical expectation is made.

Modes and local variance (c) (`shiftcharge_modes.py`). Set $\xi = x - vt$, $y = \tanh(k\xi/2)$, $u_0 = k^2(1-y^2)/2$, $h_0 = (1+y)/2$, $\omega = |q|\sqrt{1+q^2}$, $\nu = \omega - vq$. The potential mode, with asymptotic normalization $1/\sqrt{2\omega}$ and measure $\mathrm{d}q/(2\pi)$, has spatial factor

$$f_q = e^{iq\xi}\left[1 + (T_q - 1)\left(h_0 + \frac{h_0'}{iq}\right)\right], \tag{17}$$

$$T_q = -\frac{t_q - ib_q}{t_q + ib_q}, \qquad t_q = 3kq, \quad b_q = 1 - 2k^2 + 2q^2 - v\,\mathrm{sgn}(q)\sqrt{1+q^2}. \tag{18}$$

The co-moving pencil is checked directly at both signs of q and v:

$$[\partial_\xi^4 - k^2\partial_\xi^2 + 6\partial_\xi(u_0\partial_\xi) + 2iv\nu\partial_\xi - \nu^2]f_q = 0, \qquad |T_q| = 1. \tag{19}$$

Write $a_q = \Re T_q - 1$, $b_q^{\mathrm{im}} = \Im T_q$, and define

$$A_q = \tfrac{1}{2}a_q(q^2 - k^2) + kqb_q^{\mathrm{im}}, \qquad B_q = -a_q k^4/(8q^2), \tag{20}$$

$$|f_q'|^2 - q^2 = A_q(1-y^2) + B_q y^2(1-y^2)^2. \tag{21}$$

This is a direct field-mode density, not a phase-density ansatz. At a common infrared cutoff μ, the vacuum-subtracted coincident variance $g_\mu = \langle\eta_x^2\rangle/\hbar - \langle\eta_x^2\rangle_0/\hbar$ has the same two shapes with coefficients

$$A_\mu = \int_{|q|>\mu}\frac{A_q\mathrm{d}q}{4\pi\omega}, \qquad B_\mu = \int_{|q|>\mu}\frac{B_q\mathrm{d}q}{4\pi\omega}. \tag{22}$$

The limit of A_μ exists. For $0 < k < \sqrt{3}/2$, putting $R = \sqrt{1+q^2}$ gives the independently checked integral

$$A = \frac{3k^2}{4\pi}\int_1^\infty\left[\frac{1}{4R^2 + 4vR + 4v^2 - 3} + \frac{1}{4R^2 - 4vR + 4v^2 - 3}\right]\mathrm{d}R \tag{23}$$

$$= \frac{\sqrt{3}k}{4\pi}\,\mathrm{artanh}\,\frac{2k}{\sqrt{3}}. \tag{24}$$

B_μ has a soft logarithm. It must not simply be dropped in the local profile or in a momentum contraction.

Mean profile and direct Π insertion (c) (`shiftcharge_modes.py`). The order-$\hbar$ stationary field equation is

$$H_v u_1 = 3g_\mu - 2\lambda u_0, \qquad H_v = -\partial_\xi^2 + k^2 - 6u_0. \tag{25}$$

The following identities give an explicit solution, modulo translation:

$$H_v\partial_k u_0 = -2ku_0, \qquad H_v u_0'' = 6(u_0')^2, \tag{26}$$

$$u_1 = \left(-\frac{3A_\mu}{k^3} + \frac{\lambda}{k}\right)\partial_k u_0 + \frac{2B_\mu}{k^6}u_0''. \tag{27}$$

In particular the soft term is a translation-spreading contribution with zero integrated height. An adjoint check uses

$$w = \frac{1}{k^2} - \frac{3}{k^3}\partial_k u_0, \qquad H_v w = 1, \quad \int w(1-y^2)\mathrm{d}\xi = -\frac{2}{k^3}, \quad \int w y^2(1-y^2)^2\mathrm{d}\xi = 0. \tag{28}$$

Direct integration of (27), with two distinct finite values of B_μ, verifies the same mass response. Since $\Pi = \int \pi$ is linear, there is no quadratic Wick contraction of this operator. Its one-loop mean correction at fixed velocity is $\int(-vu_1)$, giving

$$\boxed{m_v = -\frac{3\sqrt{3}}{2\pi k^2}\operatorname{artanh}\frac{2k}{\sqrt{3}} + \frac{2\lambda}{k}, \qquad q_v = -vm_v.} \tag{29}$$

The latter was obtained from the stationary field insertion, independently of the energy derivative or centroid argument. Setting $q_v = 0$ just because the Gaussian contraction is zero would omit the entire mean profile.

Direct momentum and a nontrivial comparison (c) (`shiftcharge_modes.py`). The canonical momentum has both a mean-profile term $2v\int u_0 u_1$ and a quadratic fluctuation term. Define

$$J_q = \int [\Re(i f_q f_q'^*) - q]\mathrm{d}\xi = -a_q k/q + 2a_q q/k + 3b_q^{\mathrm{im}}, \tag{30}$$

$$G_q = \int (|f_q'|^2 - q^2)\mathrm{d}\xi = 4A_q/k + 8B_q/(15k). \tag{31}$$

Using $\int(\partial_k u_0)(1-y^2) = 2$ and $\int(\partial_k u_0)y^2(1-y^2)^2 = 8/45$, the soft B_q terms cancel *between* these contributions before the q integral:

$$p_v = \int \frac{\nu J_q - 2vA_q/k}{4\pi\omega}\mathrm{d}q + 2\lambda v k \tag{32}$$

$$= -\frac{2\sqrt{3}v}{\pi}\operatorname{artanh}\frac{2k}{\sqrt{3}} + 2\lambda v k. \tag{33}$$

Independent 80-digit quadratures, at five signed widths, agree with the closed forms to better than 10^{-65}. The mode pencil and profile equation are separately checked. No fitted coefficient sets the tadpole result.

(c) Only after this calculation, compare the audited conditional dispersion

$$F_g(k;\lambda) = \frac{\sqrt{3}}{\pi}(\tfrac{3}{4} - k^2)\operatorname{artanh}\frac{2k}{\sqrt{3}} - \frac{3k}{2\pi} + \frac{2\lambda k^3}{3}. \tag{34}$$

It satisfies $p_v = -\partial_v F_g$. With

$$V_1 = -p_v/P_{0,v} = \partial_P F_g, \quad \bar{m} = m_v + M_{0,v}V_1, \quad \bar{q} = -vm_v + \Pi_{0,v}V_1, \tag{35}$$

the independently computed insertion obeys (15), including $-v\bar{m}$. At $k = .31$, $\epsilon = +1$, $\lambda = 0$, for example,

$$(\bar{m}, \bar{q}, V_1) = (1.2322718175732764, -0.7212159462470162, -0.7263708360512509). \tag{36}$$

The frozen-height prediction $-2kV_1$ even has the opposite sign here. This compares two conditional one-body computations; it does not certify the missing physical boundary measure.

5 Bad/Toda calibration before the good application

Continuation and dictionary (c) /(d) (`shiftcharge_modes.py`, `shiftcharge_calibration.py`). **(c)** Use the audited formal continuation $F_b(k) = -iF_g(ik)$ with the action/Planck dictionary, including reversal of the finite quadratic coefficient under continuation. For the mass coefficient the corresponding rule is $m_b(k) = im_g(ik)$, since both M and $\hbar$ scale. Thus at fixed velocity $v = \epsilon\sqrt{1+k^2}$,

$$m_{v,b} = -\frac{3\sqrt{3}}{2\pi k^2}\arctan\frac{2k}{\sqrt{3}} - \frac{2\lambda_b}{k}, \tag{37}$$

$$p_{v,b} = -\frac{2\sqrt{3}v}{\pi}\arctan\frac{2k}{\sqrt{3}} - 2\lambda_b vk, \qquad q_{v,b} = -vm_{v,b}, \tag{38}$$

$$F_b = \frac{\sqrt{3}}{\pi}(\tfrac{3}{4}+k^2)\arctan\frac{2k}{\sqrt{3}} - \frac{3k}{2\pi} + \frac{2\lambda_b k^3}{3}, \qquad C_b = \lambda_b + 2/\pi. \tag{39}$$

These are formal continuations, not a real whole-line bad-vacuum mode sum **(d)** . The lattice-matched parameter $\lambda_b = -3/(2\pi)$ gives $C_b = 1/(2\pi)$ as in the controlling one-body audit.

(c) At reference spacing one, the established small-amplitude units are

$$\kappa = \sqrt{3}\beta, \quad k = \sinh\beta, \quad h_T = 3b\hbar_B, \quad b = e^{-1/2}, \quad P_B = \overline{P}_T/(6\sqrt{3}b), \quad E_B = \overline{E}_T/(6\sqrt{3}b^2). \tag{40}$$

The classical action obeys the corresponding scaling, so its dimensionless phase is unchanged. Put $k_1 = \rho$, $k_2 = r\rho$, $0 < r < 1$, on the right-moving overtaking branch. The same r is the leading $\sqrt{y_2/y_1}$ or cube root of the physical momentum ratio. **(d)** The additional identification tested here is between the continuum canonical *recoil representative* and the exchange-odd part of Toda's two-term *directional residual*, using their stated classical subtractions. There is no established common quantum operator/endpoint dictionary identifying those entire observables.

What actually reproduces the rational term (c) (`shiftcharge_calibration.py`). With $\sigma = +1$ for good and $\sigma = -1$ for bad, put $C = \lambda - 2\sigma/\pi$. At small amplitude the canonical insertion and Darboux coefficient obey

$$p_v = 2\sigma Ck + O(k^3), \qquad \gamma_0 = -k^{-2} + O(1), \qquad D[p] \longrightarrow -4\sigma C(r - r^{-1}). \tag{41}$$

Consequently the matched bad recoil is exactly $2(r - r^{-1})/\pi$. Independent regular quadrature verifies the audited decomposition

$$R_{12}^{(1)} = \frac{4}{\pi}\int_0^r \log(s/r)\frac{s^2}{1-s^2}\mathrm{d}s = -\frac{4}{\pi}[\chi_2(r) - r], \tag{42}$$

$$S^{(1)} = \frac{2}{\pi}(r + r^{-1}) - \pi, \tag{43}$$

$$\vartheta_N^{(1)} = R_{12}^{(1)} - S^{(1)} = \pi - \frac{4}{\pi}\chi_2(r) + \frac{2}{\pi}(r - r^{-1}), \qquad \chi_2(r) = \tfrac{1}{2}[\mathrm{Li}_2(r) - \mathrm{Li}_2(-r)]. \tag{44}$$

At $r = 1/2$, the rational part is -0.9549296585513720. The whole ϑ_N is 1.5305278132789631. Recoil accounts for neither the chi-function term nor the endpoint constant. **(c)** As an additional check against overidentification, the audited canonical continuum candidate $J + D$ differs from (44) by

$$(J + D) - \vartheta_N = -\frac{\pi}{2} - \frac{2}{\pi}\log r \log\frac{1-r}{1+r}, \tag{45}$$

a nonconstant gap, -2.0555822900856296 at $r = 1/2$. Rational agreement is not validation of the full phase.

Why this is not $c_{\rm bad}$ (c) (`shiftcharge_calibration.py`, `shiftcharge_modes.py`). The proposed finite-coefficient residual scales as

$$Q_{12} = \tfrac{4}{3} r(1-r^2)\rho^4 + O(\rho^6). \tag{46}$$

It cannot reproduce an order-one rational function of r. Conversely, every finite c disappears from this leading Toda limit, so that limit cannot determine it. Dividing the rational term by Q_{12} produces a kinematic divergence, not a coefficient in the assumed loop expansion.

(c) The actual conditional charge response makes the problem still more explicit. For either sign of the equation, for $\epsilon = +1$,

$$\bar{m} = \frac{1}{\pi k} + \left(\frac{16\sigma}{9\pi} - \frac{2\lambda}{3}\right) k + O(k^3), \tag{47}$$

$$\bar{q} = -\frac{1}{\pi k} + \left(\frac{49\sigma}{18\pi} - \frac{4\lambda}{3}\right) k + O(k^3). \tag{48}$$

Changing Π alone in T_0 would yield $4(r - r^{-1})/(3\pi)$, only two thirds of Toda's rational coefficient. The simultaneous M term in (8) yields its negative. The surviving topological substitution is

$$T_1|_P = -\frac{2\sigma}{3\pi}\rho^2(1-r^2)(r+r^{-1}) + O(\rho^4), \tag{49}$$

not a constant times Q_{12} and not (41). The leading $1/k$ charge shift is incompatible with $\Delta\Pi = \hbar\kappa P$. These are sequential one-loop coefficients; their use requires $\hbar/k^2 \ll 1$. No uniform finite-$\hbar$ zero-amplitude limit is claimed.

6 Good equation: parameter dependence and the failed frozen-height shortcut

What can be given as a function of λ (c) (all but `shiftcharge_projection.py`). Equations (29), (33) and (35) give the conditional one-body response for every finite λ. The audited small-amplitude coefficient and the isolated rational recoil are

$$C_g(\lambda) = \lambda - 2/\pi, \qquad D_g[p] \longrightarrow (-4\lambda + 8/\pi)(r - r^{-1}). \tag{50}$$

λ	C_g	Coefficient in (50)	Physical c_g
$9/(4\pi)$	$1/(4\pi)$	$-1/\pi$	undetermined
0	$-2/\pi$	$8/\pi$	undetermined

(d) These recoil coefficients are not values of c_g. They neither select a scheme nor complete the physical phase.

(c) Counterterm accounting is an independent safeguard against double counting. In the same specified chamber frame, the complete classical elastic response at fixed momenta is

$$\delta_1^{\rm ct} = \lambda K_\lambda, \qquad K_\lambda = 3\delta_0 - 4\sum_j P_j \partial_{P_j}\delta_0. \tag{51}$$

Fresh finite differences of the exact stiffness-deformed phase, with both momenta inverted anew, verify $K_\lambda^{++} = 0.05008140486841891$ and $K_\lambda^{+-} = -0.22517393868334176$ at $\alpha = (.3, .6)$. The isolated λ dependence of (50) must be accompanied by the corresponding change in the label-frame contribution; replacing the momentum convention while leaving that contribution fixed is invalid. No λK_λ already present in a representative may be added again under the name of a new CDD coefficient.

Even imposing $\bar{m} = 0$ does not rescue a universal c (c) (`shiftcharge_algebra.py`). If one nevertheless grants both the branch-velocity identification and the frozen-height assumption, then

$$\bar{q}_{\rm fr} = -M_0 F_P, \qquad c_{\rm eff}(P_1, P_2) = \frac{2}{3}\frac{F_P(P_1) - F_P(P_2)}{P_1/M_1 - P_2/M_2}. \tag{52}$$

This is a pointwise quotient, not a constant. For $\lambda = 0$ on the good branch, it is -2.4463194544 at $(k_1, k_2) = (.2, .4)$ overtaking, -4.8131022786 at $(.3, .6)$ overtaking, and -15.8575027361 at $(.2, .4)$ head-on. The corresponding three values at $\lambda = 9/(4\pi)$ are -0.7124456976, -1.5929611493, and 1.1003496931. The failure is structural, not a numerical tolerance issue.

(c) On a single right-moving branch only, its small-width limit exists. Expanding $F_P = C + D_2 k^2 + O(k^4)$ gives

$$D_2 = \frac{5\sigma\lambda}{6} - \frac{23}{9\pi}, \qquad c_{\rm fr,small} = 2D_2. \tag{53}$$

The resulting good values are $-49/(36\pi)$ at $9/(4\pi)$ and $-46/(9\pi)$ at zero; the matched bad value is $-47/(18\pi)$. They characterize this expressly false-for-the-computed-branch frozen-height shortcut, not $c_{\rm good}$ or $c_{\rm bad}$. In opposite directions even the common leading velocity correction changes sign, so it cannot be discarded as one common multiple of M in (10).

7 Ring and one collision-point projection checks

Conditional ring correction (c) (`shiftcharge_calibration.py`). For the proposed directional ring phase $\Phi(j \mid l) = \delta(j, l) + \Pi_j M_l$, the twist changes by

$$(\Pi_j M_l)_1|_P = \bar{q}_j M_{0,l} + \Pi_{0,j}\bar{m}_l. \tag{54}$$

The symmetric pair correction is exactly (9), and summing the assumed equations, including the audited self-winding term, changes the total twist by

$$(\Pi_{\rm tot} M_{\rm tot})_1 = \bar{q}_{\rm tot} M_{0,\rm tot} + \Pi_{0,\rm tot}\bar{m}_{\rm tot}. \tag{55}$$

The same charge/height response therefore appears in this algebraic check. An antisymmetric cQ_{jl} cancels from the total pair sum. **(d)** The ring holonomy is still not a quantum wavefunction-transport calculation; it provides no independent equation selecting c.

Exact two-soliton background at one point (c) (`shiftcharge_projection.py`). At $\alpha_1 = .3$, $\alpha_2 = .6$, $\epsilon_1 = \epsilon_2 = +1$, $t = X_1 = X_2 = 0$, the checker uses the exact two-soliton tau function, $\tau = 1 + E_1 + E_2 + A_{12}E_1E_2$, $E_j = e^{k_j(x - X_j) - k_j v_j t}$, and differentiates $\phi = 2\partial_x \log\tau$, $\pi = \partial_t \phi$. It evaluates the four-modulus symplectic Gram matrix directly, retaining all cross terms. Here

$$A_{12} = 0.09096862155680799, \qquad \det W = (P_{0,\alpha_1} P_{0,\alpha_2})^2 = 0.0032005493687711533. \tag{56}$$

The determinant error is below 10^{-50} at 65 digits. For the localized test fluctuation $\eta = (0, e^{-x^2})$, its symplectic projection onto these four tangents has

$$(\mathcal{P}\eta)_\phi(-\infty) = 0, \qquad (\mathcal{P}\eta)_\phi(+\infty) = -1.0756732271416573. \tag{57}$$

Thus the complementary radiation projection does not preserve fixed potential endpoints even at this particular collision point.

(d) This is an instantaneous Gram determinant, explicitly *not* the requested space-time projected fluctuation determinant. A numerical value of the latter would require a boundary-compatible evolving domain, the constraint/collective-radiation Schur complement, and time-endpoint state normalization. Those inputs are not supplied by the audited machinery. Rather than choose them by preference or assign the determinant a zero phase, the cache records `null`. A single kinematic point does not remove this missing-definition problem.

8 Claim summary, failures recorded, and conclusion

Claim and status	Evidence and limit	Checker
(c) One coefficient in the exhibited P, M wedge; defect rank zero	Polynomial primitives, mixed minors and explicit transmission lift. Not a classification.	`shiftcharge_algebra.py`
(c) Fixed-P charge and chamber algebra	Symbolic inversion, symmetric subtraction, curl counterexample, exact elastic deformation	`shiftcharge_algebra.py`
(c) Continuity/centroid implication	Regulated integration by parts; quantum domain and group-velocity identification remain conditional	`shiftcharge_algebra.py`
(c) Conditional direct m_v, q_v, p_v	Field-mode densities, pencil test, explicit mean-profile solution, independent 80-digit integrals	`shiftcharge_modes.py`
(c) Bad/Toda rational recoil match	Symbolic small-width limit and independent ratio-integral quadrature; full phase and c not matched	`shiftcharge_calibration.py`
(c) Good λ dependence	Conditional one-body responses and isolated recoil; full counterterm response checked separately	`shiftcharge_algebra.py`, `shiftcharge_modes.py`, `shiftcharge_calibration.py`
(c) Conditional ring sum and collision geometry	Charge-product expansion; direct exact-tau symplectic integration and endpoint failure	`shiftcharge_calibration.py`, `shiftcharge_projection.py`
(d) Physical residual c, projected expectation and functional determinant	Missing domain/state map and dynamical two-body calculation; explicit nulls	Open

Numerical scope (c) . The scripts use 80 decimal digits except the 65-digit collision quadrature. Analytic/integral comparisons are asserted at 10^{-65} or 10^{-70}; the finite-width Toda recoil test uses $k_1 = 10^{-16}$ and a distinct 10^{-29} truncation tolerance. Elastic-response Richardson differences use steps 10^{-14} and half that, with 10^{-50} tolerance. The collision integrates to $|x| = 600$ and separately verifies its endpoint limit. These are numerical diagnostics, not rigorous interval error bounds. All four scripts print `ALL CHECKS PASSED`; their cached runtimes total approximately 35 seconds, far below thirty minutes. Physical interpretation errors are not hidden inside those numerical tolerances.

Failures recorded.

1. **(c)** The initial stronger premise that the full residual has only one coefficient is unsupported: the defect package exhibits one wedge while explicitly retaining other unknown contributions. The work was adapted to that actual scope.

2. **(c)** The frozen-height dispersion shortcut fails constant-c and cross-channel tests; the nonzero direct height response is retained.

3. **(c)** The Gaussian linear-Π contraction alone misses the mean-profile term. The unprojected local tadpole has a soft logarithm. Its integrated mass contribution vanishes and its momentum contribution cancels by exact identities; no interval or nonzero contribution was discarded.

4. **(c)** Changing the shift charge alone gives a rational coefficient of the wrong size for Toda and omits an equal, opposite height term. The full topological substitution and $P \wedge M$ have different scaling.

5. **(c)** A Darboux-coefficient change with unchanged spatial delays fails the curl condition. The known rational recoil cannot be counted a second time as a newly computed residual.

6. **(c)** The conditional continuum/Toda full-phase comparison retains a nonconstant gap after the rational terms agree.

7. **(c)** At the single collision point the instantaneous projection fails the fixed-endpoint test. A physical two-body functional determinant and quantum ring holonomy were not evaluated, and are not reported as zero.

8. **(c)** No implementation assertion failed in these four scripts, and no tolerance was weakened. The first PDF build failed because the checker macro name conflicted with the standard LaTeX command `cal`; it was renamed before rebuilding. A second build found an accidental form-feed character in a formula command; the character was repaired and the long charge display was split for layout. The following build exposed two missing backslashes before paired left delimiters; those were restored. The two optional audit-file reads reported missing files; no claim about their unobserved verdicts is made.

Conclusion (C). **(c)** The continuity identity survives the stated boundary/domain assumptions; the conditional direct mode insertion agrees with it only after the height response is included. The tested mechanism does not isolate a universal $c(P_1M_2 - P_2M_1)$. Toda calibrates the already-known rational recoil, not that coefficient. **(d)** Therefore c_{bad} and $c_{\mathrm{good}}(\lambda)$, including its values at 0 and $9/(4\pi)$, remain undetermined. What is missing is precisely a common physical endpoint/state map, the boundary-compatible collective projection and two-body determinant (or equivalent dynamical phase input). No additional bootstrap identity, choice of the two candidate λ values, or centroid Ward identity supplied here replaces those inputs.

Internal references and reading scope. The following repository texts, rather than newly verified literature entries, supply the inherited targets and qualifications: `README.md`, `AGENTS.md`, `report.tex`, `errata.tex`; `good_twobody.tex`, `audit_good_twobody.tex` (frame and constraint passages); `defect_bootstrap.tex`, `audit_resonant_channels.tex`; `good_renorm.tex`, `onebody_scheme.tex`, `audit_onebody_scheme.tex`, `onebody_dispersion_test.tex`; `det_projected_oneloop.tex`, `audit_det_projected.tex`, `charge_frame_master.tex`, `audit_charges_action.tex`; `toda_route.tex`, `twobody_toda.tex`, `audit_twobody_toda.tex`, `audit_frame_dictionary.tex`, `audit_twobody_cont.tex`; `smx_candidates.tex`, `audit_smx.tex`. Relevant existing script implementations were read only; all reported new computations use the four independent `shiftcharge` scripts.